Wang 2019 Glioblastoma — analysis summary

RUN 7 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7 samples share ONE continuous manifold (single kNN component after island removal); per-neighbourhood iLISI rises 1.0→2.3/7 across the θ-sweep; Neftel meta-program lobes intermix, not per-sample islands; cycling a scattered ~19% minority.

Gate 2 — module geography → GO (non-branching)

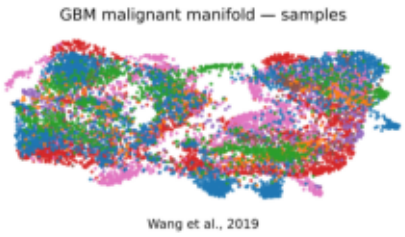
Single collapsed embryonic axis: pSom-high/Crest-low origin → Crest-high terminus (Crest decile mean pt 0.84). Gast/pSom co-vary (+0.16), both oppose Crest (−0.87/−0.49). Non-branching case — GO with Gate 3 = NA.

Gate 3 — hub cross-repression → NA

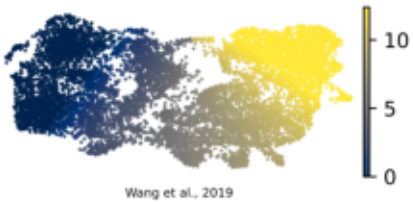
No embryonic-module bifurcation: PPT fork count is resolution-dependent, not stable — harmony-space 1,3,2,1 (Nodes 20/30/50/100), UMAP-space 1,2,2,2,3 (Nodes 20/30/50/80/120); no tip is Gast-positive (tips are pSom-vs-Crest poles). Node-cluster Gast–pSom r=+0.07 sits inside the label-shuffle null [−0.35,+0.28]. Neftel AC/OPC/NPC/MES multipolarity ≠ module bifurcation — no branch to test.

STEP FIGURES

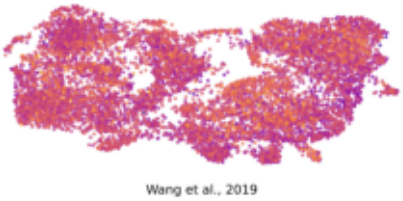
Manifold — samples



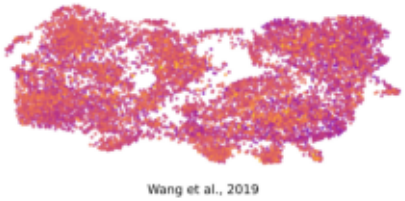
Pseudotime



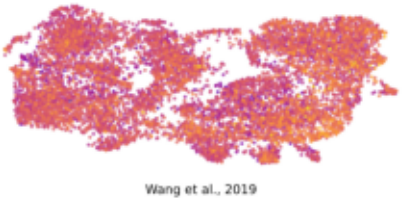
Gastrulation



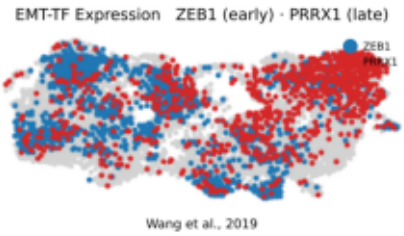
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 9,632 malignant cells, 7 samples (of 10). Malignant cell-state manifold, embryonic-program re-analysis (CCCA meta-atlas).

Verdict.
The embryonic pattern reproduces as a single collapsed axis — a pSom/Gast-high origin grading to a Crest-high terminus — NOT an embryonic bifurcation. Neftel GBM state-multipolarity does not carry a Gast↔pSom branch that survives a fold-stability + independence null. Gate 1 GO / Gate 2 GO / Gate 3 NA. Geometry linear (non-branching). EMT-TFs ZEB1 (early) → PRRX1 (late). Module means Gast −0.003 / pSom +0.008 / Crest −0.007.

Parameter choices

Sample threshold	237 malignant (largest multiplicative count-gap 396→143); 7/10 samples, 9,632 cells	**Embedding**	n_pcs=20; Harmony θ=10 on sample; spread=1.3 (PC1≈PC2≈11%, 2D cloud)
Cell cycle	scored de novo (Tirosh S/G2M absent in source); orthogonal to modules (|r|≤0.12)	**Trajectory**	scFates curve (non-branching), Nodes=20 μ=100; fold-free (SA 2.82, DC1 corr +0.98)
Root	Gast/pSom-high, Crest-low tip (Crest late/terminal)	**EMT-TFs**	ZEB1 (70%) & PRRX1 (40%); VIM excluded (marker); Snail/Twist/FoxC undetectable